

The main findings and Implications

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- Multimodal fusion of 11 satellite and gridded Earth observation streams predicts county-level life expectancy across 3,084 contiguous-US counties over 20 years (2000-2019) with MAE = 1.08 +/- 0.02 yr ($R^2 = 0.631 \pm 0.013$) under strict county-grouped cross-validation, using no census or sociodemographic input.
- Nighttime land surface temperature features carry approximately 6x the cumulative SHAP attribution of all daytime channels combined, with a protective model response below a minimum overnight temperature of about 7.5 degrees C that transitions to a penalty above roughly 10-12 degrees C.

Implications of the main findings

- The chronic nocturnal thermal environment - not acute daytime heat - emerges as the principal satellite-observable signal associated with county longevity, providing an annually updatable, radiometric criterion for prioritising residential cooling assistance, nocturnal heat-alert systems, and urban-greening choices that preserve radiative cooling at night.
- Because all input streams are globally available and continuously updated, the framework is extensible to regions where civil registration and vital statistics coverage is incomplete, supporting satellite-based monitoring of UN Sustainable Development Goal Target 3.9.